NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

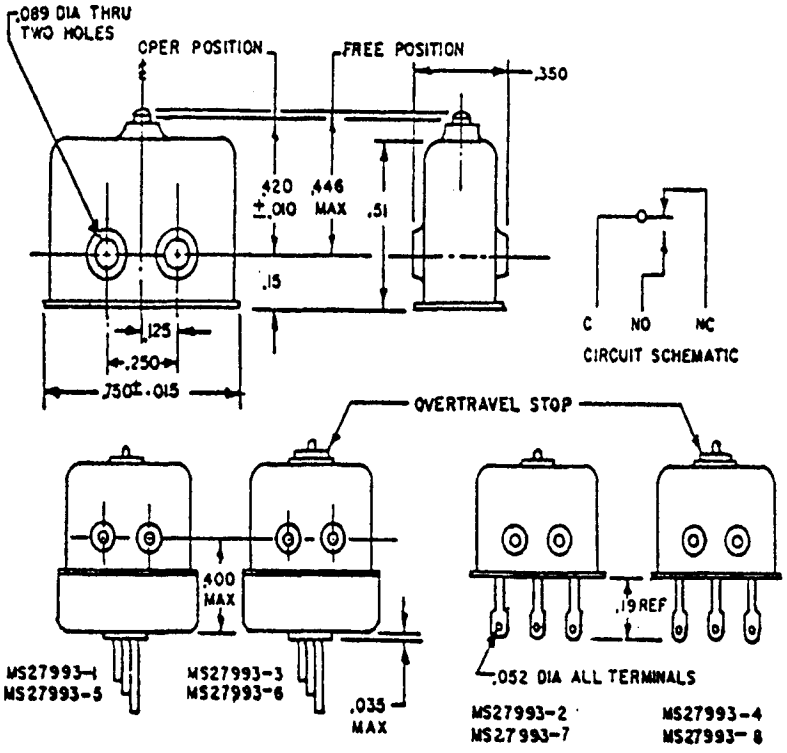
MIL-PRF-8805/45E
3 July 1974
SUPERSEDING
MIL-S-8805/45D
18 April 1972

PERFORMANCE SPECIFICATION SHEET

SWITCHES, SENSITIVE, SPDT, 5 AMPERES, RESILIENT SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the switches described herein shall consist of this document and the latest issue of Specification MIL-S-8805.



Inches	mm
.005	.13
.010	.25
.015	.38
.02	.51
.035	.89
.052	1.32
.089	2.26
.125	3.18
.15	3.81
.19	4.83
.250	6.35
.350	8.89
.400	10.16
.420	10.67
.446	11.33
.51	12.95
.750	19.05

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.02 (.51 mm) for 2 decimals and ±.005 (.13 mm) for 3 place decimals.
3. Switch configuration optional provided dimensions specified are not exceeded.
4. Insulated conductors in accordance with M5086/1-22-*, 18 inches long.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

(E) denotes changes

FSC 5930

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REQUIREMENTS:

Dimensions and configurations	See figure
Enclosure design.	4 (Resilient).
Temperature characteristics	
MS27993-1·	1 (-55° to +85°C).
MS27993-2·	(-55° to +125°C)
MS27993-3·	1 (-55° to +85°C).
MS27993-4·	(-55° to +125°C).
MS27993-5·	1 (-55° to +85°C).
MS27993-6	1 (-55° to +85°C).
MS27993-7:	(-55° to +125°C).
MS27993-8	(-55° to +125°C).
Shock type:	M (100G, test condition I, method 213 of MIL-STD-202)
Vibration grade	2 (10 to 2,000 Hertz (Hz)).
Operating characteristics	See table I.
Contact resistance·	80 milliohms maximum.
Dielectric withstanding voltage·	
Sea level:	1,000 Vrms
Seal test·	Only watertight test shall be performed in group "A."
Ⓔ Mechanical endurance	
MS27993-1 thru -4	100,000 cycles at a rate of 10 to 12 cycles per minute
MS27993-5 thru -8	100,000 cycles
Electrical endurance	50,000 cycles
Electrical rating	See table II
Military part numbers.	See table I.
Quality assurance:	Group B inspection - See table IV

TABLE I. Part numbers and operating characteristics.

Part number	Actuating Ⓔ force	Releasing force	Movement differential	Over-travel	Weight	Weight with wire	Applied actuator force	Termination
MS27993-1	22 oz max	6 oz min	.0005 to .005 inch	.007 inch max 1/	---	.0616 lbs max	Does not apply	Lead-wire
MS27993-2	22 oz max	6 oz min	.0005 to .005 inch	.007 inch max 1/	.02 lbs max	---	Does not apply	Solder
MS27993-3	22 oz max	6 oz min	.0005 to .005 inch	.006 inch min	---	.0616 lbs max	10 lbs max	Lead-wire
MS27993-4	22 oz max	6 oz min	.0005 to .005 inch	.006 inch min	.02 lbs max	---	10 lbs max	Solder
MS27993-5	10 oz max	3 oz min	.0005 to .005 inch	.007 inch max 1/	---	0616 lbs max	Does not apply	Lead-wire
MS27993-6	10 oz max	3 oz min	.0005 to .005 inch	.006 inch min	---	.0616 lbs max	10 lbs max	Lead-wire
MS27993-7	10 oz max	3 oz min	.0005 to .005 inch	.007 inch max 1/	.02 lbs max	---	Does not apply	Solder
MS27993-8	10 oz max	3 oz min	.0005 to 005 inch	.006 inch min	.02 lbs max	---	10 lbs max	Solder

Ⓔ 1/ Overtravel stop not provided. In use, do not exceed.

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TABLE II Electrical ratings.

Load	Sea Level		50,000 feet
	28 Vdc	115 Vac, 400 Hz	28 Vdc
	Amperes	Amperes	Amperes
Resistive	5	5	5
Inductive	3	3	3
Lamp	1.5	1.5	1.5

Qualification.

Group submission: See table III.

TABLE III. Qualification group submission.

Examination or test	Basic switch	Other switch samples	Extent of approval
Qualification inspection table of MIL-S-8805 groups I thru VIII, inclusive	MS27993-8 (All sample units)	MS27993-1, -2, -3, -4, -5, -6, -7, (3 sample units) Visual and mechanical examination	All

TABLE IV. Group B inspection.

Examination or test	Basic switch	Other switch samples
Group B inspection table of MIL-S-8805 group I thru IV	MS27993-8 (All sample units)	MS27993-1 MS27993-2 MS27993-3 MS27993-4 MS27993-5 MS27993-6 MS27993-7 Visual and mechanical examination

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - SL
Navy - OS, SH
Air Force - 17, 80, 85
DSA - ES

User activities:
Army - MI, AV, WC
Navy - MC, AS

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5930-0962)

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   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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